POMDP Motion Planning Algorithm Based on Multi-Modal Driving Intention

Lin Li, Wanzhong Zhao, and Chunyan Wang

***Abstract*—On highways, the interaction with surrounding vehicles is very crucial for the decision-making and planning of autonomous vehicles. However, the multi-modal driving intentions of surrounding vehicles have brought great challenges. Aiming at the multi-modal driving intention of surrounding vehicles, a multi-modal driving risk field based on dynamic collision region is proposed, and multi-modal driving intention partially observable markov decision process (MDI-POMDP) decision framework is established, which integrating behavior decision and motion planning. Firstly, the multi-modal probability distribution of driving intention is fused to establish a driving risk field. Moreover, combined with the longitudinal safety distance model and lateral driving direction, the concept of dynamic collision area is proposed in the driving risk field. Then, MDI-POMDP is formulated to analyze the influence of the uncertainty on planning, which is caused by the multi-modal driving intention of surrounding vehicles. In the following, with the help of the previous state, a time-dependent deep reinforcement learning (DRL) algorithm recurrent deterministic policy gradient (RDPG) is designed to enhance the current observation, to solve the optimal driving policy under partial observation and generate the optimal trajectory. Furthermore, the simulation results show that the performance of our proposed motion planning algorithm is outstanding, compared with the states-of-the-art methods. And our algorithm has the powerful ability to model the multi-modality of driving intention, to ensure the traffic safety.**

***Index Terms*—Deep reinforcement learning (DRL), multi-modal driving intention (MDI), partial observed Markov decision process (POMDP), recurrent deterministic policy gradient (RDPG).**

I. Introduction

AUTONOMOUS Vehicles (AV) have emerged as one of the most promising means to solve serious traffic problems such as collision accidents, congestions and pollutions. Extensive researches have been done on perception, planning, and control, of course a range of significant progress has been made, among where motion planning module is at the heart of Autonomous Driving (AD) [1], [2], [3]. In the field of AV, motion planning refers to three main functions, including prediction, decision-making and trajectory generation. The prediction step provides future evolution of traffic situations, and allows for risk estimation that is fundamental to react to widely variable situations. Considering the possible future motions and interaction with surrounding vehicles, the best behavior is selected out of pre-defined high-level motions in decision-making step. And then, a feasible, safe, dynamical and comfortable trajectory is generated in final step, taking into account all kinds of driving criteria and vehicle constraints. In recent research, it is generally recognized that multiple possible future situations and coupling interactions make motion planning remain an important challenge. To be more specific, our paper focuses on the motion planning on highway driving, due to its simple driving structures and the driver's limited behaviors, making it the most reachable context for the first fully autonomous driving. However, the behavior that surrounding vehicle will take at next time is not certain but multiple, as shown in Fig. 1. The probability distribution of surrounding vehicle motion is multi-modal. Considering that the driving speed is high, if the multi-modality of surrounding vehicle is ignored, the serious collision will occur. In terms of risk assessment, current research is limited to the simple driving scenarios like car-following. It is difficult to apply to the complex scenarios, where the driving intention is multi-modal.

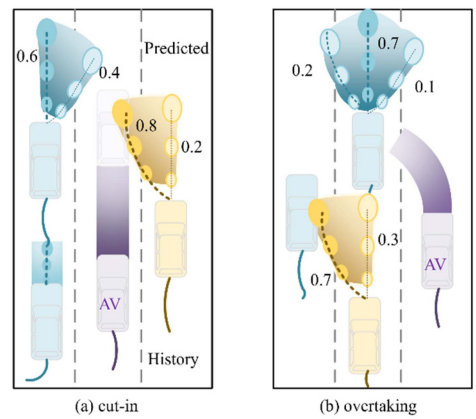

Fig. 1. Typical multi-modal driving scenarios on highway.

A considerable amount of literature has been published on motion planning on highways, using a wide variety of approaches such as different optimization techniques, model predictive control methods, artificial intelligence, and machine learning etc. Inspired by the robot motion planning, lattice presentation algorithm, such as A∗, D∗, and RRT [4], [5], are easy to extend to the autonomous vehicles. However, the planning of vehicles

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is largely more complex than robots, due to the dynamic traffic environments and the collision safety constraints. In order to meet both vehicle dynamics and safety constraints, some researchers use MPC [6] and other optimal control methods [7], [8], integrated with Bezier [9] or clothoid curves [10], to generate feasible and smooth trajectories. Whereas these methods may face with high computational complexity, optimal algorithms like Particles Swarm Optimization (PSO) [11] are more efficient to deal with solving problem. Though these traditional methods are easily applicable, they are hard to consider the interaction information during planning. Because the number of surrounding vehicles states is large and variable, the cost of computation become very high, which is difficult to handle. More recently, artificial intelligence has made significant progress and provides a renew method for motion planning. Considerable research has attempted to formulate the planning problems based on Deep Learning, Markov Decision Process and Reinforcement Learning [12], [13] etc. Note that the interaction is hard to model explicitly [14], Deep Q-learning and Graph Attention Network is integrated in [15] to achieve scalable and proactive driving. To deal with the issue of computational complexity and robustness of traditional motion planning methods for autonomous driving, an end-to-end motion planning model based on a deep cascaded neural network is proposed in [16]. A combined approach is also proposed in [17], where the sampled scenarios from the POMDP can directly be transferred to the MPC problem and information from the POMDP solution can be used to judiciously select the important scenarios. The integration allows for the long planning horizons and larger action spaces. To plan with continuous action space, Soft Actor-Critic is used in ref [18] to solve the motion planning problem. Parallel Planning [19] is proposed to deal with emergency situations, where a hybrid generative model including a variational auto-encoder (VAE) and a generative adversarial network (GAN) is utilized to learn from virtual emergencies generated in artificial traffic scenes.

It can be seen that Deep Learning has powerful ability to model traffic situation and POMDP is good at uncertainty, and their integration with Deep Reinforcement Learning show great potential to deal with the motion planning problems. However, most of the existing research only focus on the sensor uncertainty, and fail to notice the prediction uncertainty. Taking into account the multi-modal driving intention, the planning module must deal with the uncertain traffic states. Especially when the uncertainty coupled with other vehicles' interactions, the problems become even complex. In this paper, the uncertainty caused by multimodal driving intention is considered in the designed decision making model. However, this way may bring some troubles in the solving process. To remedy it, an advanced learning algorithm that could introduce previous states is proposed, so that the optimal driving action could be solved in uncertain observation.

Aiming at the multi-modality of driving intention of surrounding vehicles, a multi-modal driving risk field based on dynamic collision region is proposed, and MDI-POMDP decision framework is established, which integrates decision-making and motion planning. The main contributions are given as follows:

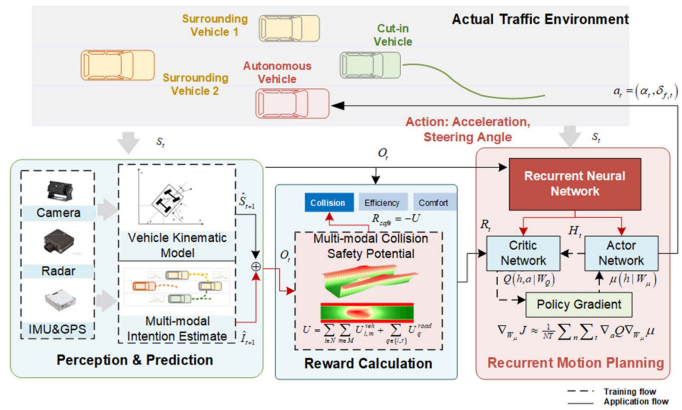

Fig. 2. Proposed motion planning framework.

1) To fully estimate the collision risk, the multimodal distribution of driving intention is fused to establish the driving risk field. Also, the concept of dynamic collision area is proposed, where longitudinal safety distance model and lateral driving direction are combined. 2) MDI-POMDP is established to investigate the influence of the multi-modal driving intention on the decision-making. 3) Moreover, under uncertain observation, a time-dependent DRL algorithm RDPG is designed to solve the optimal driving action and generate the optimal trajectory, which is realized by enhancing the current observation with the previous state.

The rest of this article is organized as follows. In Section II, the overall motion planning system is introduced. Then, the multi-modal driving risk potential based on dynamic collision area is designed in Section III. The MDI-POMDP motion planning model is proposed in Section IV and the optimal algorithm is presented in Section V. In Section VI, experiments results are illustrated. Finally, conclusions are drawn in Section VII.

II. System Overview

In order to deal with multi-modality during decision-making process, multi-modal collision safety potential and POMDP decision framework are proposed. As shown in Fig. 2, the whole framework mainly is composed of three parts, including perception and prediction module, Reward Calculation module, and Recurrent Motion planning module. The perception and prediction module consists of a sensor system, including Camera, Radar and IMU, which are used to perceive actual traffic state S_t. The camera and radar equipped to autonomous vehicles could be used to track surrounding vehicles, and IMU & GPS are used to get the motion states of autonomous vehicles. Then these data are gathered and sent to the prediction part, to predict future evolution of the traffic environment. When ego vehicle receives an action a_t, the future motion states are predicted based on vehicle kinematic model and an intention inference model is introduced to estimate multi-modal distribution of surrounding vehicle's driving intention. According to the observed traffic states, the reward calculation module is designed to assess current states and chosen driving actions. In particular, consider multiple possible driving intention of surrounding vehicles, a

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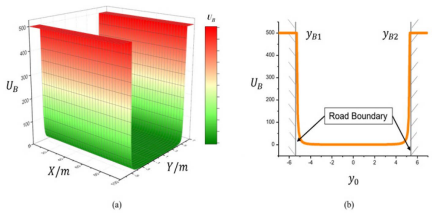

Fig. 3. Road boundary potential field.

multi-modal collision safety potential is invented to fully assess the collision risk in multi-lane scenarios. In the following, the multi-modal and uncertain driving intention is modeled in the POMDP framework, called Multi-modal Driving Intention POMDP. In addition, the Recurrent Deterministic Policy Gradient is designed to solve this partial observed planning problem.

III. Dynamic Collision-Area Based Multi-Modal Driving Risk Potential

Road safety, which involves the life of passengers and property of vehicles, is the most important decision-making indicator during the driving decision-making [20], [21]. However, on the highways, the vehicle speed is relatively high, and the body shape is very different. It is hard to ensure the safety and feasibility of the planned path by simply limiting the distance between the vehicle centroids. In addition, considering the multi-lane scene, the multi-modality of driving intention of surrounding vehicles, as well as the dynamic acceleration and heading angle of vehicles, this paper integrates the multi-modal distribution of driving intention to establish a multi-modal driving risk field based on the dynamic collision area, so as to adapt to the complex traffic scene on highways.

The driving risk field can be regarded as the possibility that a target vehicle and its surrounding objects could collide at a certain position, which is determined by their interaction. This means that the closer the surrounding objects are, the greater the collision possibility, and the higher the potential energy. On highways, it is also necessary to consider the restriction of road boundary during driving. Therefore, the driving risk field is generated by the weighted superposition of the surrounding vehicles and the road boundary,

$$U_{total} = \omega_B U_B + \omega_V U_V \tag{1}$$

A. Road Boundaries Potential Field

On highways, the vehicle is not allowed to drive away from its current boundaries. As illustrated in Fig. 3, the road boundary potential field U_B is generated in part of the road boundary line and extends to infinity with the vehicle approaching gradually.

$$U_{Bq} = \begin{cases} \eta/d_{Bq}^2, d_{Bq} \le d_{B,\min} \\ \eta/d_{B,\min}^2, \text{others} \end{cases} \tag{2}$$

$$d_{Bq} = y_{Bq} - y_0 \tag{3}$$

where η is the coefficient of the road boundary potential field. d_{Bq} is the lateral distance between the autonomous vehicle and road boundary y_{Bq} and q denotes the left or right boundary. $d_{B,\min}$ is the minimum measurable distance, denote as 0.01 m in this paper.

B. Obstacle Vehicle Potential Field

As for the vehicle potential field, multiple possible driving intentions are first considered to improve the collision risk assessment ability. According to the estimated probability of driving intentions, for each vehicle V_i, the potential energy of each possible driving intention is added by probability,

$$U_{vi} = \lambda M_i \sum_k \Pr\left(m_k^i\right) \exp\left(-k_r \hat{d}_{0,i,k}\right) \tag{4}$$

$$M_i = \sigma_i \times \left(1.566 \times 10^{-14} \times v_i^{6.687} + 0.3345\right) \tag{5}$$

where λ, k_r are the coefficients of vehicle potential and relative distance between obstacle and ego vehicle, respectively. M_i is the mass after velocity correction [22] and σ_i is the physical mass of the vehicle V_i. $Pr(m_k^i)$ represents the probability of the k-th driving intention for vehicle V_i. $\hat{d}_{0,i,k}$ means the distance between autonomous vehicle V_0 and vehicle V_i when the vehicle V_i executes the driving intention m_k^i.

1) Dynamic Collision Area: In the traditional Artificial Potential Field model, the vehicle is regarded as a static point, and the collision risk is evaluated by calculating the distance from the surrounding obstacles to ego vehicle centroid, which ignores the shape, size and dynamic characteristics of the vehicle. More importantly, the existing Artificial Potential Field model does not consider the change of lateral driving direction. However, on highways, it is common for adjacent vehicles to change the driving direction and insert into our original lane, that is main reason for accidents. Therefore, combining the longitudinal safety distance model and lateral driving direction, the concept of dynamic collision area is proposed in this paper. And then the distance to this dynamic collision area is used for vehicle potential field.

According to the shape of vehicles, the collision area is defined as an ellipse. On highways, the moving speed of the vehicles is high. Therefore, the long and short axes of collision ellipse are defined as the longitudinal and lateral safety distances [23],

$$X_s = X_{th} + v\cos\theta T_0 + 0.5 * \Delta(v\cos\theta)^2/a_n \tag{6}$$

$$Y_s = Y_{th} + v\sin\theta T_0 + 0.5 * \Delta(v\sin\theta)^2/a_n \tag{7}$$

where X_{th}, Y_{th} are respectively longitudinal and lateral minimum clearance distance. T_0 is the safe time gap. Δ is the different with previous time step. a_n is the maximum deceleration.

In addition to the dynamic characteristic, the vehicle pose is also considered in our potential field, which is represented by heading angle θ. The vehicle collision boundary equation can be represented by a rotated ellipse,

$$\frac{((x-x_i)\cos\theta_i + (y-y_i)\sin\theta_i)^2}{X_s^2} + \frac{((x_i-x)\sin\theta_i + (y-y_i)\cos\theta_i)^2}{Y_s^2} = \frac{1}{2} \tag{8}$$

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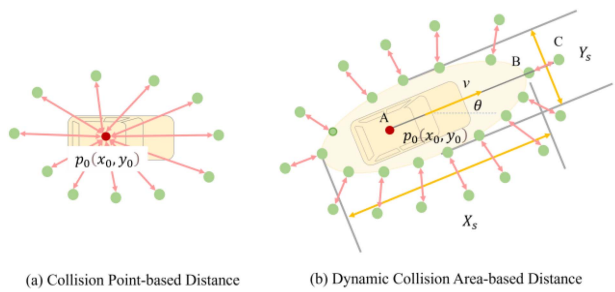

Fig. 4. The difference between traditional point-based and our dynamic area-based distance.

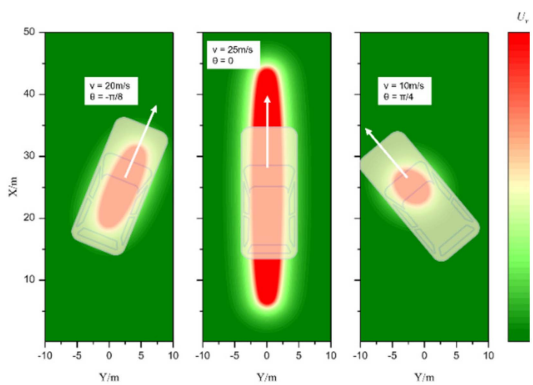

Fig. 5. Obstacle vehicle potential field for different driving behaviors.

where x, y are respectively longitudinal and lateral positions around the obstacle vehicles. x_i, y_i are the positions of the obstacle vehicle. θ_i is the heading angle of the obstacle vehicle.

2) Distance to Collision Area: Define the minimum distance from position (m, n) to the collision boundary as the distance to collision area [24]. The distance can be defined as follows,

$$\hat{d}_{0,i} = \min_{(x,y)\in E} \sqrt{(x-m)^2 + (y-n)^2}$$
$$= \sqrt{(m-x_i)^2+(n-y_i)^2}\,\frac{\sqrt{2Y_{i,s}^2 p_i^2 + 2X_{i,s}^2 q_i^2} - X_{i,s}Y_{i,s}}{\sqrt{2Y_{i,s}^2 p_i^2 + 2X_{i,s}^2 q_i^2}} \tag{9}$$

$$\begin{cases} p_i = (m-x_i)\cos\theta_i + (n-y_i)\sin\theta_i \\ q_i = (m-x_i)\sin\theta_i + (n-y_i)\cos\theta_i \end{cases} \tag{10}$$

where x, y are positions on the collision ellipse. The difference between traditional point-based and our dynamic area-based distance is illustrated in Fig. 4. It can be seen that our potential could consider the shape of the vehicle, which provides more reliable safety constraint.

In Fig. 5, the obstacle vehicle potential fields regarding to different velocity and directions are presented. The higher is the velocity, the longer is the long-axis of the collision ellipse. And the collision ellipse could be rotated with the driving direction. It can be seen that the proposed risk potential is able to describe the collision potential more precise. Then, the different modal vehicle potential fields are fused according to the probability distribution of driving intention, shown in Fig. 6.

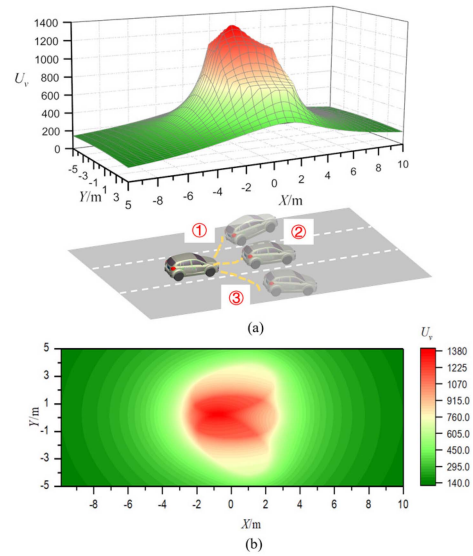

Fig. 6. Integrated potential field considering multi-modal driving intention.

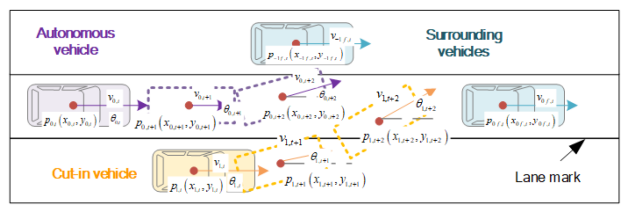

Fig. 7. Driving scenario description.

IV. MDI-POMDP Motion Planning Model

Due to multi-modality of driving intention of surrounding vehicles, the traffic states at decision stage are uncertain and are not fully observable. In this paper, Multi-modal Driving Intention Partial Observable Markov Decision Process (MDI-POMDP) [25] is formulated, where the traffic states are represented by a set of state vectors and action vectors. Executing these vectors, the motions of autonomous vehicles could be controlled. Especially, a set of observation states and observation model are introduced to represent the estimation of surrounding vehicles.

A. Formulation of MDI-POMDP

A POMDP model can be formulized as a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{O}, \mathcal{Z}, \gamma)$. $\mathcal{S}$ is state space, $\mathcal{A}$ denotes the action space and $\mathcal{Z}$ is the observation space. The conditional function $\mathcal{T}(s_{t+\Delta t}|s_t, a_t)$ represents state transition probabilities to state $s_{t+\Delta t} \in \mathcal{S}$ with an action $a \in A$ in the state $s_t \in \mathcal{S}$. The observation function $\mathcal{O}$ models the probability of observing $z \in \mathcal{S}$, when the state is $s_{t+\Delta t} \in \mathcal{S}$ and an action $a \in A$ is taken. The reward function $R(s, a)$ returns an immediate reward and $\gamma \in (0, 1]$ is the discount factor.

1) Traffic States: According to the property of POMDP, the state space $\mathcal{S}$ must hold sufficient information for either decision making or belief update. In our defined driving scenarios in Fig. 7, the state vector $s_t \in \mathcal{S} \subseteq \mathbb{R}^{D_s}$, D_s is the dimension of the state vector, includes the states of the autonomous vehicle

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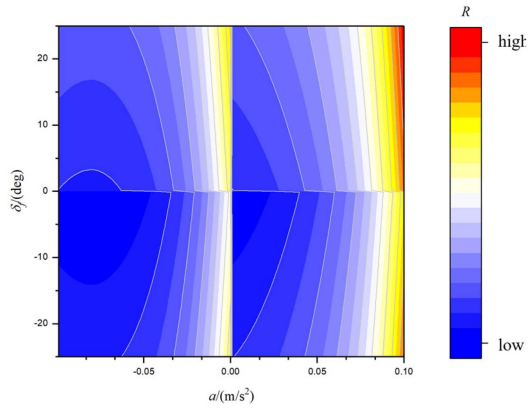

Fig. 8. The reward map within action space.

and surrounding vehicles, which can be denoted as the following Eq. (11),

$$s_t = \left[s_t^{V_0}, s_t^{V_1}, \ldots, s_t^{V_i}, \ldots, s_t^{V_N}\right]^T \tag{11}$$

where $s_t^{V_0}$ is the state of autonomous vehicle, which in this case comprises the vehicle position $(x_t^{V_0}, y_t^{V_0})$, velocity $v_t^{V_0}$ and heading $\theta_t^{V_0}$. $s_t^{v_i}, i \in \{1, 2, \ldots, N\}$ is the state of other vehicles and *N* is the number of other vehicles involved. For surrounding vehicles, at time *t*, except for vehicle motion states, multimodal distribution of driving intention $m_t^{V_i}$ needs to be covered by the surrounding vehicle state $s_t^{V_i}$ for proper state reward calculating.

$$s_t^{V_0} = \left[x_t^{V_0}, y_t^{V_0}, v_t^{V_0}, \theta_t^{V_0}\right]^T \tag{12}$$

$$s_t^{V_i} = \left[x_t^{V_i}, y_t^{V_i}, v_t^{V_i}, \theta_t^{V_i}, m_t^{V_i}\right]^T \tag{13}$$

$$m_t^{V_i} \in \mathcal{M} = \{LK, LLC, RLC\} \tag{14}$$

where *LK* means the *i*-th vehicle has the intention of keeping motion; *LLC* means the vehicle turn left to change lane; *RLC* represents right lane-changing.

2) Driving Actions and Motion Transition Model: In order to integrate the decision-making and motion planning layers, our driving action is designed as specific operating metric variables. Our action space is a two-dimensional and continuous, including acceleration α_t and steering wheel angle δ_t at time *t*. The action chosen by the autonomous vehicle could be represented as,

$$a_t = (\alpha_t, \delta_t) \tag{15}$$

where the acceleration α_t varies in the continuous range from -3 to 2 m/s^2. And the steering wheel angle δ_t ranges from $-15°$ to $15°$, which is also a continuous space.

The motion transition model updates the traffic states given the previous state and driving actions. In our scenarios, the state transition process involves state update of autonomous vehicle and surrounding vehicles.

$$\mathcal{T}\left(s_{t+\Delta t}|s_t, a_t\right) = \Pr\left(s_{t+\Delta t}^{V_0}|s_t^{V_0}, a_t^{V_0}\right) \prod_{i=1}^{N} \Pr\left(s_{t+\Delta t}^{V_i}|s_t^{V_i}\right) \tag{16}$$

TABLE I
ACTION VALUES OF DIFFERENT DRIVING INTENTIONS

	LK	LLC	RLC
α	α	0	0
δ	0	-15°	15°

where $Pr(s_{t+\Delta t}^{V_0}|s_t^{V_0}, a_t^{V_0})$ represent the evolution of involved vehicles from motion states at initial time $s_t^{v_0}$. $a_t^{v_0}$ is the planned action for autonomous vehicle by proposed model.

For surrounding vehicles, $a_t^{v_i}$ is estimated by the probability distribution of driving intention $m_t^{v_i}$, which is updated according our observation model.

$$\Pr\left(s_{t+\Delta t}^{V_i}|s_t^{V_i}\right) = \sum_{m_t^{V_i}} \Pr\left(s_{t+\Delta t}^{V_i}|s_t^{V_i}, a_t^{V_i}\right) \Pr\left(a_t^{V_i}|s_t^{V_i}, m_t^{V_i}\right) \tag{17}$$

where $Pr(s_{t+\Delta t}^{V_i}|s_t^{V_i}, a_t^{V_i})$ is the motion states transition function of surrounding vehicle V_i, and the $Pr(a_t^{V_i}|s_t^{V_i}, m_t^{V_i})$ is the action estimation function of surrounding vehicle V_i.

To be specific, the action value a_m for each driving intention is predefined, as list in Table I. And then the final action is calculated as the weighted sum of the predefined values and probability distribution of driving intention.

$$a_t^{V_i} = \sum_{\mathcal{M}} a_m \Pr\left(m_t^{V_i}|s_t^{V_i}\right) \tag{18}$$

When received an action $a_t^{V_i} = (\alpha_t^{V_i}, \delta_t^{V_i}), i \in \{01, 2 \ldots, N\}$ at time *t*, the next motion state of vehicle V_i could be described by vehicle kinematics equation,

$$\begin{cases} x_{t+\Delta t} = x_t + v_{t+\Delta t}\cos\left(\theta_{t+\Delta t}\right)\Delta t \\ y_{t+\Delta t} = y_t + v_{t+\Delta t}\sin\left(\theta_{t+\Delta t}\right)\Delta t \\ v_{t+\Delta t} = v_t + \alpha_t \Delta t \\ \theta_{t+\Delta t} = \theta_t + v_{t+\Delta t}\tan\left(\delta_t\right)/l \end{cases} \tag{19}$$

where Δt is the planning horizon. l is the distance between shafts.

3) Observation Space and Observation Model: A set of observations are added to the model, and the traffic states are no longer directly obtained. Instead, an observation results are provided, which give us a hint about what states the system is in. At the same time, the observation model is also defined, which tells us the probability of the observation of the traffic states in the decision model.

Similar to state space, the observation space $\mathcal{Z}$ is defined as,

$$z_t = \left[z_t^{V_0}, z_t^{V_1}, \ldots, z_t^{V_i}, \ldots, z_t^{V_N}\right]^T \tag{20}$$

where $z_t^{V_0}$ and $z_t^{V_i}, i \in \{12, \ldots, N\}$ are respectively motion states of autonomous vehicle and surrounding vehicles observed from autonomous vehicle. For ego vehicle, $z_t^{V_0}$ is a noisy measurement of the true motion states, and the observation model is described as,

$$\begin{aligned} z_{t+\Delta t}^{V_0} &= \hat{s}_{t+\Delta t}^{V_0} \sim \Pr\left(z_{t+\Delta t}^{V_0}|s_t^{V_0}, a_t^{V_0}\right) \\ &= N\left(\mu_x^{V_0}, \mu_y^{V_0}, \mu_v^{V_0}, \mu_\theta^{V_0}, \sum_{x,y,v,\theta}^{V_0}\right) \end{aligned} \tag{21}$$

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where $\hat{s}_{t+\Delta t}^{V_0}$ is the observed motion states of ego vehicle V_0. $Pr(z_{t+\Delta t}^{V_0}|s_t^{V_0}, a_t^{V_0})$ represents the observation function of the vehicle V_0. $\mu_x^{V_0}$, $\mu_y^{V_0}, \mu_v^{V_0}, \mu_\theta^{V_0}$ are respectively the means of positions, velocity and heading angle. $\Sigma_{x,y,v,\theta}^{V_0}$ is the covariance matrix of these states.

As for surrounding vehicles, the motion states also could be viewed as true metrics with sensor noise. However, the driving intentions of them are not able to observed directly, which only could be obtained by estimation. And the observation model for surrounding vehicles is given by,

$$z_{t+\Delta t}^{V_i} = \hat{s}_{t+\Delta t}^{V_i} \sim \Pr\left(z_{t+\Delta t}^{V_i}|s_t^{V_i}, a_t^{V_i}\right)$$
$$= \Pr\left(m_t^{V_i}|s_t\right) N\left(\mu_x^{V_i}, \mu_y^{Vi}, \mu_v^{V_i}, \mu_\theta^{V_i}, \sum_{x,y,v,\theta}^{V_i}\right) \quad (22)$$

In terms of driving intention, the probability distribution $\Pr(m_t^{V_i}|s_t)$ is estimated according to all the involved vehicle states s_t, and the detailed model has been proposed in our previous publication, referring to [26].

4) Instant Driving Reward: The reward function is responsible to guide the driving action of autonomous vehicle and achieve the driving goals. In highway scenarios, three main objectives are desired to be satisfied, including collision safety R_{safe}, efficiency R_{eff}, and comfort R_{com}. Thus, the whole reward is given by,

$$R(s,a) = \omega_{safe} R_{safe} + \omega_{eff} R_{eff} + \omega_{com} R_{com} \quad (23)$$

where the $\omega_{safe}, \omega_{eff}, \omega_{com}$ are weight coefficients, and they could be adjusted to various driving preferences.

The traditional safety indicators [27], [28] are hard to correctly assess the collision risk, due to multi-modal driving intention and coupling longitudinal and lateral motion states of the surrounding vehicles, particularly for the cut-in vehicles. To overcome this limitation, the dynamic collision-area based driving risk potential (proposed in Section II) is used to evaluate the collision risk level, which includes the potential collision with surrounding vehicles and the road boundary.

$$R_{safe,t} = -U_{total,t} = -\left(\omega_B U_{B,t} + \omega_V U_{V,t}\right) \quad (24)$$

The vehicles always expect to reach the destination with least time. Therefore, on the premise of safety, the decision-making system should also be encouraged to improve the velocity as much as possible. In this paper, the efficiency reward is given as the deviation between real velocity and desired velocity, where the sigmoid function is selected. The closer to the desired velocity, the higher the efficiency reward.

$$R_{eff,t} = 1/\left(1 + \exp\left(-\kappa * \left(v_t^{V_0} - v_d\right)\right)\right) \quad (25)$$

where $v_t^{V_0}$ is the actual velocity of ego vehicle V_0. v_d is the desired velocity which usually is given by the maximum within regulation limitations. κ is the adjustment coefficient of velocity.

Considering that the more stable the vehicle is, the higher the ride comfort of passengers is. Therefore, the change value of heading angle is used to quantify the driving comfort. The comfort payoff function is defined as follows,

$$R_{com,t} = \exp\left(-m * abs\left(\dot{\theta}_t^{V_0}\right)\right) \quad (26)$$

where $\dot{\theta}_t^{V_0}$ is the yaw rate of the autonomous vehicle V_0. m is the adjustment coefficients of yaw rate. In Fig. 8, it seems that the acceleration has more impact on the reward, and the higher acceleration means the higher reward.

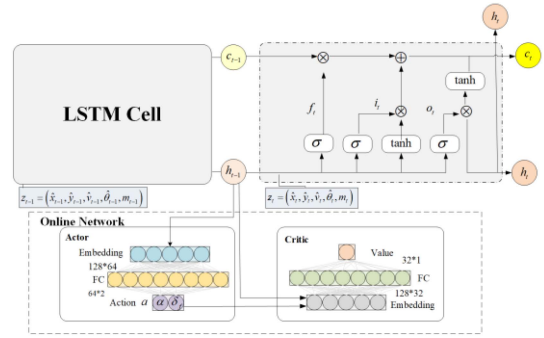

Fig. 9. The network architecture of DRL algorithm.

V. RDPG-Based Optimal Driving Policy Solving

The decision model has been established in the previous section, where the traffic states can be observed, and the system evolution and reward after a chosen driving action have been described. In this section, we need to design a learning algorithm to solve the problem that how to acquire the optimal driving policy, which means how to drive according to current traffic states.

Considering the noises of sensors and uncertainty of the observation models, the current observations are hard to be reliable to make decisions. In our algorithm, there needs to be a way to represent previous actions and observations, which helps to infer actual current states. The emergence of recurrent neural networks makes it possible for autonomous vehicles to synthesize the historical states and features [29]. Recurrent Deterministic Policy Gradient makes use of this characteristics and combines it with Deterministic Policy Gradient, to solve the partial observable problems during our learning process.

The network architecture used in this paper can be seen in Fig. 9 Take the historical actions and observations as input, which is defined as,

$$h_t = [a_0, z_1, a_1, z_2, \ldots, a_{t-1}, z_t] \quad (27)$$

Then encode the current observations z_t and previous states vectors h_{t-1} into the hidden state vectors h_t via a recurrent LSTM network [30],

$$h_t = LSTM\left(z_t, h_{t-1}; W_{LSTM}\right) \quad (28)$$

where W_{LSTM} represents the network parameters of LSTM.

The features of all observations and actions in the previous sequence have been stored in the hidden state h_{t-1} in the LSTM, therefore each time, only we need to do is adding observations at current time.

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A. Memory Actor-Critic

Similar to Deep Deterministic Policy Gradient (DDPG) [31], RDPG is also based on the Actor-Critic framework. While the different is that a temporal memory unit is added in RDPG. The whole algorithm framework consists of two main modules, namely actor and critic modules.

The actor module is composed of a strategy network, which is used to select actions, according to the hidden states h_t.

$$a_t = \text{Actor}\left(h_t\right) = \mu\left(h_t|W_\mu\right) \tag{29}$$

where μ is a deterministic mapping, usually is a fully-connected network and W_μ represents the parameters of strategy network.

After each action selection, the critic module evaluates the new state to assess whether long-term returns are better or worse than expected. The critic module is composed of an estimated value neural network *u*, which is a state-action value function.

$$Q_t = \text{Critic}\left(h_t, a_t\right) = u\left(h_t, a_t|W_u\right) \tag{30}$$

where *u* is also a fully-connected network and W_u represents the parameters of value networks.

The critic module could self-adjust according to historical information h_t and real-time feedback *R*, and then updates and guides the whole strategy network.

B. Value-Based Optimal Parameter Learning Process

Ego vehicle tries to act in a specific traffic state, and evaluates its consequences, according to the instant rewards it receives and estimated value of its state. By repeatedly trying all actions in all states, the driving experience pool could be obtained,

$$\boldsymbol{M} = \left\{\left(z, a, R, z'\right)_m\right\}^M \tag{31}$$

From that the vehicle can judge the overall best driving behavior through long-term discount rewards.

Our RDPG continues the idea of fixed target networks in DQN, and each network is subdivided into target networks (denoted as $\hat{\mu}, \hat{u}$) and online networks (denoted as $\mu,\ u$). However, the update of the target network is somewhat different.

According to the Bellman optimal equation, the state value of an optimal strategy must be equal to the expected return of the optimal behavior in that state. Thus, value function is estimated by iterative updating based on Bellman equation [32].

$$Q^*_{n+1}\left(h, a\right) = \mathbb{E}\left[R + \gamma \max_{a'} \hat{Q}^*\left(h', a'\right)|h, a\right] \tag{32}$$

where Q^*_{n+1} represents the optimal value after n+1 iteration, which is produced by online strategy network *u*. $\hat{Q}^*$ represents the optimal value produced by target network $\hat{u}$. $h',\ a'$ are respectively next hidden state and action after *h*, *a*.

This value iteration algorithm could make value network u converge to the optimal value function,

$$Q^*\left(h, a\right) \approx u\left(h, a|W_u\right) \tag{33}$$

through minimizing the loss function L_t that changes at each iteration time *t*,

$$L_t\left(W_u\right) = \mathbb{E}_{h,a\sim\mu(\cdot)}\left[\left(y_t - u\left(h_t, a_t|W_u\right)\right)^2\right] \tag{34}$$

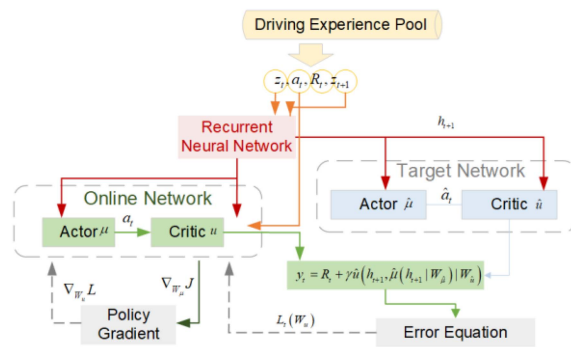

Fig. 10. The optimal parameter updating process of our RDPG.

where $y_t = \mathbb{E}_{h\sim\varepsilon}\left[R + \gamma \max_{a_{t+1}} \hat{u}(h_{t+1}, a_{t+1}; W_{\hat{u}})|h_t, a_t\right]$ is the objective at iteration time *t*. When optimizing the loss function L_t, temporarily fix the parameter $W_{\hat{u}}$ at previous iteration time t+1. Monte Carlo [33] and temporal difference learning [34] are two effective methods to solve the optimal action. In our planning process, the whole state sequence is hard to obtain. Thus, for calculating the TD gradient of small batches *M* selected from the driving experience pool, that is, the error gradient of the value network,

$$\begin{aligned}&\nabla_{W_u} L\left(W_u\right)\\ &\approx \frac{1}{MT}\sum_m\sum_t\left(u\left(h_{m,t}, a_{n,t}|W_u\right) - y_t^m\right)\\ &\times \nabla_{Wu} u\left(h_{m,t}, a_{m,t}|W_u\right)\end{aligned} \tag{35}$$

where only the states and actions of next time step are involved. y_t^m is the expected objective value by target value network $\hat{u}$ for the batch *m*. The input of the target value network is the hidden state $h_{m,t+1}$ and driving action $a_{m,t+1}$ at the next time. This action is produced by the target strategy network $\hat{\mu}$, whose input is $h_{m,t+1}$. The calculation process of y_t^m can be described as the following equation,

$$y_t^m = R_t^m + \gamma \hat{u}\left(h_{m,t+1}, \hat{\mu}\left(h_{m,t+1}|W_{\hat{\mu}}\right)|W_{\hat{u}}\right) \tag{36}$$

And then, Adam optimizer could be used to update the value network parameters. In terms of the update of strategy networks, the gradient is calculated based on the gradient of value network,

$$\begin{aligned}&\nabla_{W_\mu} J\\ &\approx \frac{1}{MT}\sum_m\sum_t \nabla_a u\left(h, a|W_u\right)|_{h=h_{m,t}, a=\mu(h_{m,t})}\\ &\times \nabla_{W_\mu}\mu\left(h|W_\mu\right)|_{h=h_{m,t}}\end{aligned} \tag{37}$$

Based on the above update algorithm, the parameters of the whole networks are updated by sampling for numerous times in the driving experience pool, until the long-term discount return converges to the ideal optimal value. The flow chart of the whole learning algorithm is shown in the Fig. 10.

VI. Experiments and Results

The main purpose of the experiments is to verify the efficacy of the proposed learning algorithm and driving performance

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TABLE II
HYPER-PARAMETER SETTINGS OF DIFFERENT RLS

	DDPG	PG	RDPG	MDI-RDPG
Actor learning rate	0.001	0.01	0.001	0.001
Critic learning rate	0.01	/	0.001	0.001
Gamma	0.9	0.95	0.98	0.98
Numbers of nodes in actor	30	10	64	64
Numbers of nodes in critic	30	/	32	32
Numbers of LSTM nodes	/	/	128	128

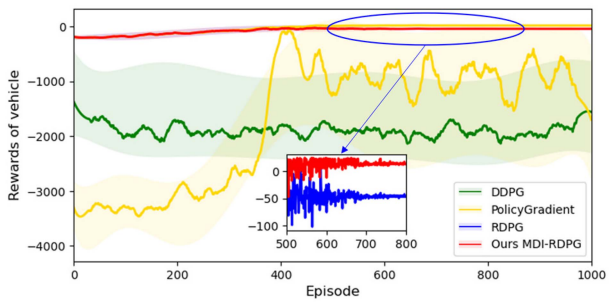

Fig. 11. The reward curves of different algorithms.

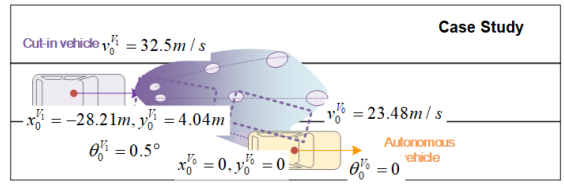

Fig. 12. The experiment scenarios description.

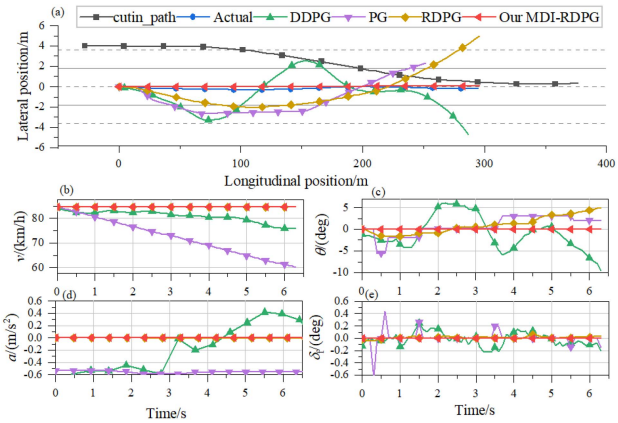

Fig. 13. The experiment results. (a) Planned path. (b) Velocity. (c) Heading angle. (d) Acceleration. (e) Front wheel angle.

tested in real traffic scenarios. All following used algorithms are implemented in TensorFlow (TF) on HP computer (RAM: 8GB, processor: Intel Core i7-6700, Operating frequency: 3.40 GHz).

A. Learning Performance of Different RLs

Considering our driving actions are continuous, state-of-the-art RL methods that are suitable for continuous actions are used as benchmarks. Policy Gradient (PG) and Deep Deterministic Policy Gradient (DDPG) are two main typical RL learning algorithms [35], which are chosen as comparisons in our simulation. PG is based on error backpropagation to increase the proper action within continuous spaces. While DDPG is a baseline with our proposed algorithm, which has a similar structure. The different is that a temporal recurrent unit is added in our proposed algorithm, so that the historical traffic states could be integrated with current observations. The details of the parameter settings are list in the Table II.

The episodic average rewards of different algorithms are presented in Fig. 11. It is observed that our algorithm is extremely superior to all comparisons in learning performance. The initial points of these algorithms are randomly selected. In this case, the initial reward of our algorithm is dramatically higher than the others. For this reason, the upward of our reward is not very obvious and our reward only takes about 250 episodes to reach the highest point. At about episode 700, our reward curve is even stable and converges to a satisfying reward. DDPG receives second highest reward at initial training step, though during the training process the reward seems steadily stable. As for PG reward, the initial reward is lowest, however, it rises in the first 400 training episodes. And then the reward fluctuates around about -1000 in the following episodes. It is concluded that our proposed algorithm has the best efficiency, astringency and well-applicable for decision-making problems.

B. Driving Scenario Tests

Then, the simulations of well-trained RL motion planning controller are conducted to illustrate the driving performance. To be more persuasive, two typical real driving scenarios are selected from public driving datasets. Also, the function of our proposed driving risk potential will be assessed in this part.

Case Study: The first simulation driving scenario is selected from HighD dataset [36]. HighD is an authoritative highway dataset collected by institute of automotive engineering of Aachen University of technology, on German roads around Cologne, where all kinds of driving behaviors are included. Since the traffic density of collected area is relatively low, the scenario is constructed with two vehicles, as shown in Fig. 12, which is a typical cut-in scenario. Our task is to consider how to plan the motion of the yellow vehicle, when perceived the adjacent vehicle wants to change lanes into its own lane. The initial positions of them are set as $p_0 = (0, 0), p_1 = (-28.21, 4.04)$. The initial velocities are set as $v_0 = 23.48$ m/s, $v_1 = 32.5$ m/s respectively. And for the autonomous vehicle, the driving intention of the purple vehicle is unknown and multi-modal.

The planning results of different algorithms are presented in Fig. 13. As shown in Fig. 13(a) it could be observed that our planned path is best closet to the actual path. When the PG and DDPG planning controller notice the collision risk of the cut in vehicle, they tend to overreact and give a great steering. As a result, their paths are both from one lane to another lane. With regard to our RDPG path, it shows a steady pattern and is more smooth. It maybe for the reason that our RDPG could integrate

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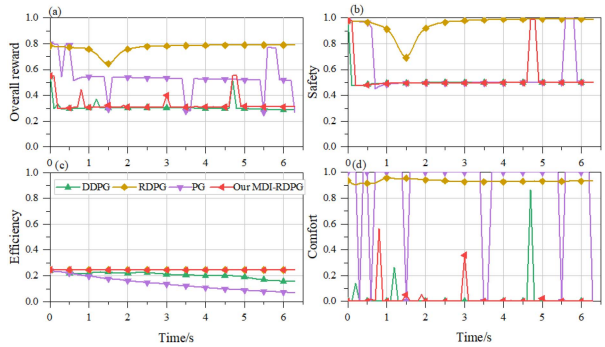

Fig. 14. The rewards of different algorithms. (a) Overall reward. (b) Safety reward. (c) Efficiency reward. (d) Comfort reward.

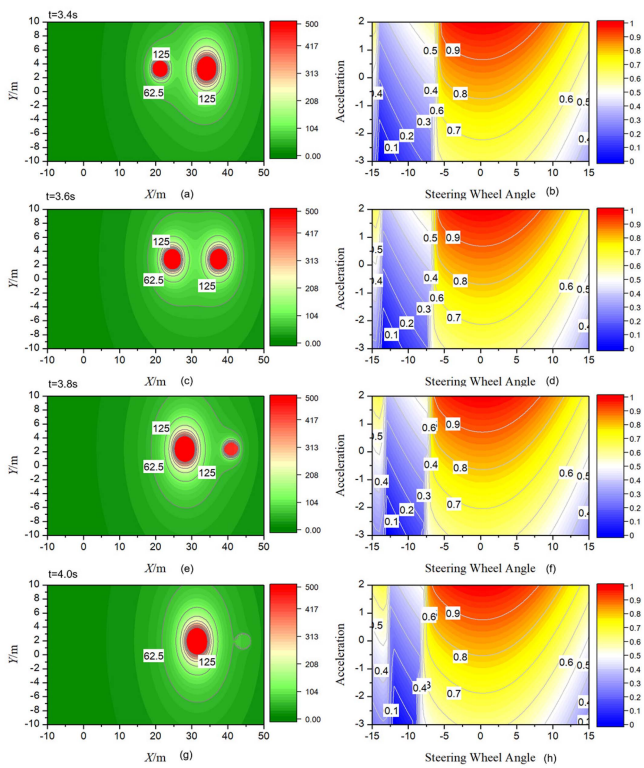

Fig. 15. The driving risk potentials and reward maps at different time steps during planning process. (a) t = 3.4 s. (b) t = 3.6 s. (c) t = 3.8 s. (d) t = 4.0 s.

the historical states, and it is more confidential to maintain current driving states until it judges that it has to operate to avoid the obstacle vehicle. The detailed motion states during planning process are shown in Fig. 13(b) and (c), which are respectively velocity, heading angle. The last two figures, Fig. 13(d) and (e), show the planned control inputs for the ego vehicle. Meanwhile, all kinds of driving rewards considered in our models are calculated to further demonstrate the driving performance, shown in Fig. 14(a) and (b). Overall, it is unsurprising that our RDPG achieves the advantageous driving rewards, including safety, efficiency, and comfort. Even RDPG changes it motion later, just decelerate at about 200 meters, and it gets a better indicator in terms of safety. In conclusion, not only with regard to the safety, or the efficiency but also the comfort, our proposed algorithm has achieved the extraordinary performance.

Furthermore, in order to illustrate the performance of our proposed multi-modal driving risk potential, the potential fields at different time steps are provided in Fig. 15. The times when the driving behavior changes are chosen for testing the multi-modality on the driving risk. At $t = 3.4$ s, the obstacle vehicle is more likely to keep its own lane, 80%, and has 19% probability to left lane-change. Therefore, the potential has two peaks, and the point around the lane-keeping behavior is higher and the collision area is broader. Regarding to the reward map, the highest reward occurs at the largest acceleration and left turning around 2°. From $t = 3.6$ s, the probabilities of the lane-keeping and left lane-changing begin to be equal, and the collision areas of these two behaviors become equal. The reward map is nearly same with the last time. At $t = 3.8$ s, the obstacle vehicle tends to change lanes, so that the peak point is the behind one. In the reward map, the acceleration is not recommended and turning is more encouraged. The situation at t = 4.0 s is in the similar case.

VII. Conclusion

In this paper, a motion planning framework considering multi-modality of surrounding vehicle intentions is proposed. First, a new driving risk potential is established by fusing the multi-modal probability distribution of driving intention. Moreover, the driving risk potential is based on the dynamic collision area, which combined with longitudinal safe distance and lateral driving direction. Secondly, the MDI-POMDP framework is formulated, where the uncertainty caused by the multi-modal driving intention is model in the planning process. Then, with the help of knowledge of previous states, a time-dependent recurrent DPG is designed to solve optimal driving policy under the partial observed situation. Finally, the experiments are conducted, and other state-of-the-art methods are compared in our experiments. It is found that our method is best in terms of the collision safety, path smooth, efficiency, and even the comfort. Furthermore, the multi-modality of driving intention could be explicitly modeled in our planning model, which helps to achieve more reliable and safe driving.

However, our paper only deals with one surrounding vehicle and in reality, there often exists multi vehicles. In future, multi-vehicles will be involved in our research, and the focuses on surrounding vehicles should be discriminated according to traffic situations.

References

[1] J. Ma and F. Wu, "Feudal multi-agent deep reinforcement learning for traffic signal control," in *Proc. 19th Int. Conf. Auton. Agents Multiagent Syst.*, 2020, pp. 816–824.

[2] K. Muhammad, A. Ullah, J. Lloret, J. Del Ser, and V. H. C. De Albuquerque, "Deep learning for safe autonomous driving: Current challenges and future directions," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4316–4336, Jul. 2021, doi: 10.1109/TITS.2020.3032227.

[3] J. Rios-Torres and A. A. Malikopoulos, "A survey on the coordination of connected and automated vehicles at intersections and merging at highway on-ramps," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 5, pp. 1066–1077, May 2017, doi: 10.1109/TITS.2016.2600504.

[4] Y. Kuwata, J. Teo, G. Fiore, S. Karaman, E. Frazzoli, and J. P. How, "Real-time motion planning with applications to autonomous urban driving," *IEEE Trans. Control Syst. Technol.*, vol. 17, no. 5, pp. 1105–1118, Sep. 2009, doi: 10.1109/TCST.2008.2012116.

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[5] S. Xu et al., "System and experiments of model-driven motion planning and control for autonomous vehicles," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. 52, no. 9, pp. 5975–5988, Sep. 2022, doi: 10.1109/TSMC.2021.3131141.

[6] J. Changhao, H. Miaohua, and S. Liyang, "An autonomous vehicle motion planning method based on dynamic programming," in *Proc. 17th Int. Comput. Conf. Wavelet Act. Media Technol. Inf. Process.*, 2020, pp. 394–398, doi: 10.1109/ICCWAMTIP51612.2020.9317333.

[7] J. Muller, J. Strohbeck, M. Herrmann, and M. Buchholz, "Motion planning for connected automated vehicles at occluded intersections with infrastructure sensors," *IEEE Trans. Intell. Transp. Syst.*, to be published, doi: 10.1109/TITS.2022.3152628.

[8] J. Chen, W. Zhan, and M. Tomizuka, "Autonomous driving motion planning with constrained iterative LQR," *IEEE Trans. Intell. Veh.*, vol. 4, no. 2, pp. 244–254, Jun. 2019, doi: 10.1109/TIV.2019.2904385.

[9] J. Nan, W. Deng, and B. Zheng, "Intention prediction and mixed strategy nash equilibrium-based decision-making framework for autonomous driving in uncontrolled intersection," *IEEE Trans. Veh. Technol.*, to be published, doi: 10.1109/TVT.2022.3186976.

[10] X. Jin, Z. Yan, G. Yin, S. Li, and C. Wei, "An adaptive motion planning technique for on-road autonomous driving," *IEEE Access*, vol. 9, pp. 2655–2664, 2021, doi: 10.1109/ACCESS.2020.3047385.

[11] D. Kim, G. Kim, H. Kim, and K. Huh, "A hierarchical motion planning framework for autonomous driving in structured highway environments," *IEEE Access*, vol. 10, pp. 20102–20117, 2022, doi: 10.1109/ACCESS.2022.3152187.

[12] S. Kuutti, R. Bowden, Y. Jin, P. Barber, and S. Fallah, "A survey of deep learning applications to autonomous vehicle control," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 2, pp. 712–733, Feb. 2021.

[13] B. R. Kiran et al., "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022, doi: 10.1109/TITS.2021.3054625.

[14] W. Ding, L. Zhang, J. Chen, and S. Shen, "EPSILON: An efficient planning system for automated vehicles in highly interactive environments," *IEEE Trans. Robot.*, vol. 38, no. 2, pp. 1118–1138, Apr. 2022, doi: 10.1109/TRO.2021.3104254.

[15] P. Cai, H. Wang, Y. Sun, and M. Liu, "DQ-GAT: Towards safe and efficient autonomous driving with deep Q-learning and graph attention networks," *IEEE Trans. Intell. Transp. Syst.*, 2022, doi: 10.1109/TITS.2022.3184990.

[16] S. Song, X. Hu, J. Yu, L. Bai, and L. Chen, "Learning a deep motion planning model for autonomous driving," in *Proc. IEEE Intell. Veh. Symp.*, 2018, pp. 1137–1142, doi: 10.1109/IVS.2018.8500703.

[17] C. H. Ulfsjöö and D. Axehill, "On integrating POMDP and scenario MPC for planning under uncertainty–with applications to highway driving," in *Proc. IEEE Intell. Veh. Symp.*, 2022, pp. 1152–1160.

[18] X. Tang, B. Huang, T. Liu, and X. Lin, "Highway decision-making and motion planning for autonomous driving via soft actor-critic," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 4706–4717, May 2022, doi: 10.1109/TVT.2022.3151651.

[19] L. Chen, X. Hu, W. Tian, H. Wang, D. Cao, and F. Y. Wang, "Parallel planning: A new motion planning framework for autonomous driving," *IEEE/CAA J. Automatica Sinica*, vol. 6, no. 1, pp. 236–246, Jan. 2019, doi: 10.1109/JAS.2018.7511186.

[20] M. Aramrattana, T. Larsson, C. Englund, J. Jansson, and A. Nåbo, "A novel risk indicator for cut-in situations," in *Proc. IEEE 23rd Int. Conf. Intell. Transp. Syst.*, 2020, pp. 1–6, doi: 10.1109/ITSC45102.2020.9294315.

[21] J. Wurts, J. L. Stein, and T. Ersal, "Collision imminent steering at high speeds on curved roads using one-level nonlinear model predictive control," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 8278–8289, 2020, doi: 10.1109/ACCESS.2021.3063795.

[22] J. Wang, J. Wu, and Y. Li, "The driving safety field based on driver–vehicle–road interactions," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 4, pp. 2203–2214, Aug. 2015, doi: 10.1109/TITS.2015.2401837.

[23] M. Xu, Y. Luo, G. Yang, W. Kong, and K. Li, "Dynamic cooperative automated lane-change maneuver based on minimum safety spacing model," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 1537–1544, doi: 10.1109/ITSC.2019.8917095.

[24] Y. Tian, H. Pei, J. Yang, J. Hu, Y. Zhang, and X. Pei, "An improved model of driving risk field for connected and automated vehicles," in *Proc. IEEE Int. Intell. Transp. Syst. Conf.*, 2021, pp. 285–291, doi: 10.1109/ITSC48978.2021.9565069.

[25] D. Deng, C. Wang, and W. Wang, "Joint air-to-ground scheduling in UAV-aided vehicular communication: A DRL approach with partial observations," *IEEE Commun. Lett.*, vol. 26, no. 7, pp. 1628–1632, Jul. 2022, doi: 10.1109/lcomm.2022.3167110.

[26] L. Li, W. Zhao, C. Wang, Q. Chen, and F. Chen, "BRAM-ED : Vehicle trajectory prediction considering the change of driving behavior," *IEEE/ASME Trans. Mechatronics*, to be published, doi: 10.1109/TMECH.2022.3188011.

[27] F. Luis and G. Moncayo, "Safe following distance analysis for traffic cellular automata modeling," in *Proc. IEEE 6th Int. Conf. Eng. Technol. Appl. Sci.*, 2019, pp. 1–6.

[28] D. Iberraken, L. Adouane, and D. Denis, "Safe autonomous overtaking maneuver based on inter-vehicular distance prediction and multi-level bayesian decision-making," in *Proc. IEEE 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 3259–3265, doi: 10.1109/ITSC.2018.8569401.

[29] L. Lei, Y. Tan, G. Dahlenburg, W. Xiang, and K. Zheng, "Dynamic energy dispatch based on deep reinforcement learning in IoT-driven smart isolated microgrids," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 7938–7953, May 2021, doi: 10.1109/JIOT.2020.3042007.

[30] H. Ge, Z. Yan, W. Yu, and L. Sun, "An attention mechanism based convolutional LSTM network for video action recognition," *Multimedia Tools Appl.*, vol. 78, no. 14, pp. 20533–20556, 2019, doi: 10.1007/s11042-019-7404-z.

[31] Y. O. Anisimov and D. A. Katcai, "Deep reinforcement learning approach for autonomous driving," *J. Phys. Conf. Ser.*, vol. 1864, no. 1, pp. 1–18, 2021, doi: 10.1088/1742-6596/1864/1/012013.

[32] Q. Zheng, K. Zheng, H. Zhang, and V. C. M. Leung, "Delay-optimal virtualized radio resource scheduling in software-defined vehicular networks via stochastic learning," *IEEE Trans. Veh. Technol.*, vol. 65, no. 10, pp. 7857–7867, Oct. 2016, doi: 10.1109/TVT.2016.2538461.

[33] S. Mo, X. Pei, and C. Wu, "Safe reinforcement learning for autonomous vehicle using monte carlo tree search," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6766–6773, Jul. 2022, doi: 10.1109/TITS.2021.3061627.

[34] X. Shen, X. Zhang, and Y. Wang, "Kernel temporal difference based reinforcement learning for brain machine interfaces," in *Proc. IEEE Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2021, pp. 6721–6724, doi: 10.1109/EMBC46164.2021.9631086.

[35] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv: 1707.06347*.

[36] R. Krajewski, J. Bock, L. Kloeker, and L. Eckstein, "The highD dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems," in *Proc. 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 2118–2125, doi: 10.1109/ITSC.2018.8569552.

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